

Co-2 Assessment tool-3

Case study

Case Study 1 – Retail Sales Performance Dashboard Q1)

Requirement	Suitable Visualization	Justification
Regional revenue comparison	Bar Plot	A bar plot clearly compares revenue across different regions, making it easy to identify the highest-performing region.
Monthly sales trend	Line Chart	A line chart effectively displays changes in sales over time and highlights

Requirement	Suitable Visualization	Justification
Product category contribution	Stacked Bar Chart	increasing or decreasing trends. A stacked bar chart shows the contribution of each product category to the total revenue while allowing comparison among categories.
Advertising expenditure vs Sales	Scatter Plot	A scatter plot is the best choice to visualize the relationship between advertising expenditure and sales and identify correlation.

Requirement	Suitable Visualization	Justification
Regional sales distribution	Histogram	A histogram shows how sales values are distributed across different ranges, helping understand the spread of regional sales.

Q2) i. R code for bar plot:

```
library(ggplot2)
```

```
retail <- data.frame(
```

```
  Region = c("North","South","East","West","Central"),
```

```
  Revenue = c(214000,235000,131000,155000,148000)
```

```
)
```

```
ggplot(retail, aes(x=Region, y=Revenue, fill=Region))
```

```
+
```

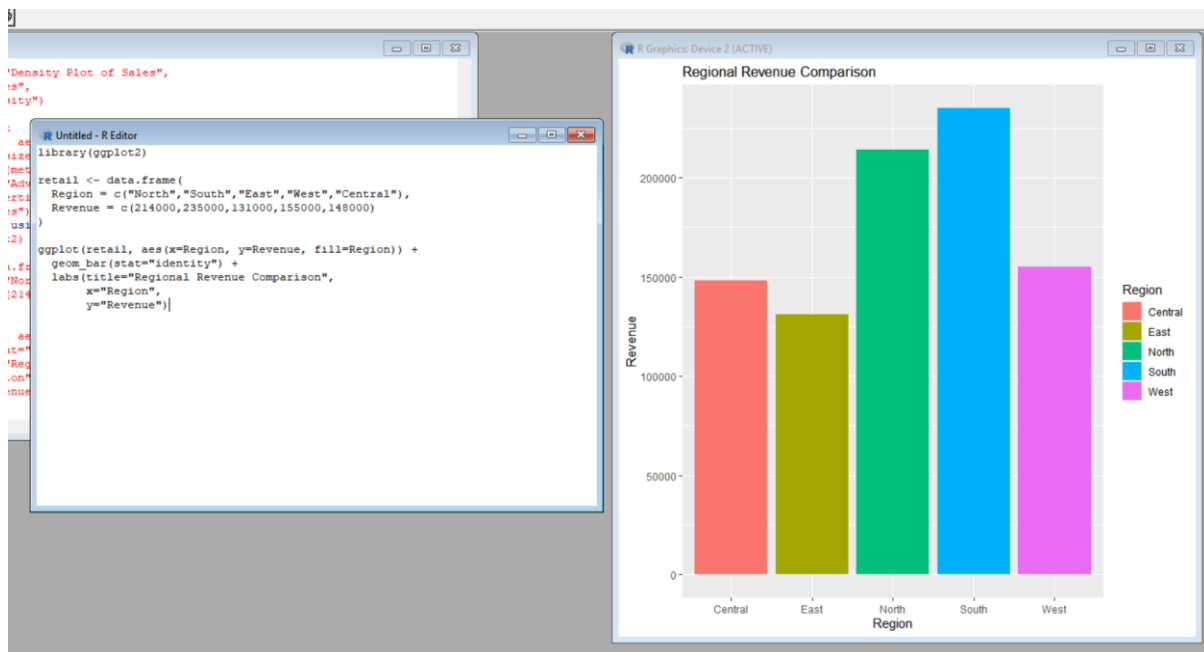
```
  geom_bar(stat="identity") +
```

```
  labs(title="Regional Revenue Comparison",
```

```
        x="Region",
```

y="Revenue")

Output:



ii. R code for stacked bar chart:

```
library(ggplot2)
```

```
retail <- data.frame(
```

```
  Region =
```

```
  c("North", "North", "South", "South", "East", "East", "West",  
    "West", "Central", "Central"),
```

```
  Category =
```

```
  c("Electronics", "Furniture", "Electronics", "Clothing", "Furniture",  
    "Clothing", "Electronics", "Furniture", "Clothing", "Electronics"),
```

```

Revenue =
c(120000,94000,130000,105000,70000,61000,90000,65
000,78000,70000)
)

ggplot(retail, aes(x=Region, y=Revenue,
fill=Category)) +

  geom_bar(stat="identity") +

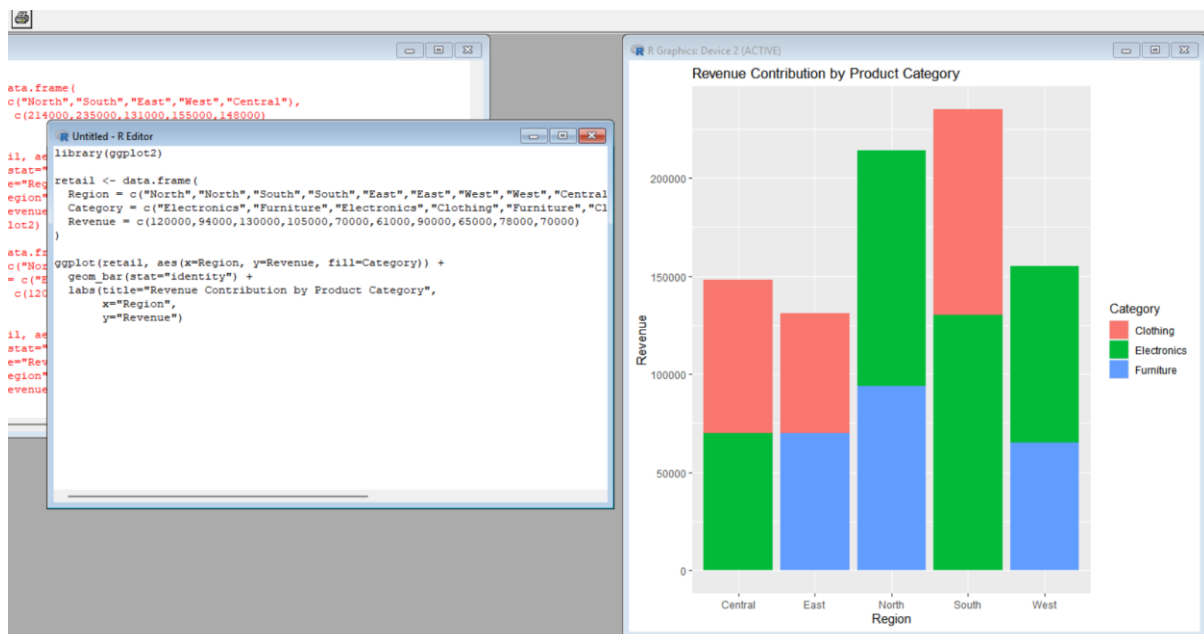
  labs(title="Revenue Contribution by Product
Category",

        x="Region",

        y="Revenue")

```

Output:



iii. R code for histogram:

```
library(ggplot2)
```

```
retail <- data.frame(
```

```
  Sales =
```

```
c(500,620,580,700,650,720,760,690,810,780,850,900)
```

```
)
```

```
ggplot(retail, aes(x=Sales)) +
```

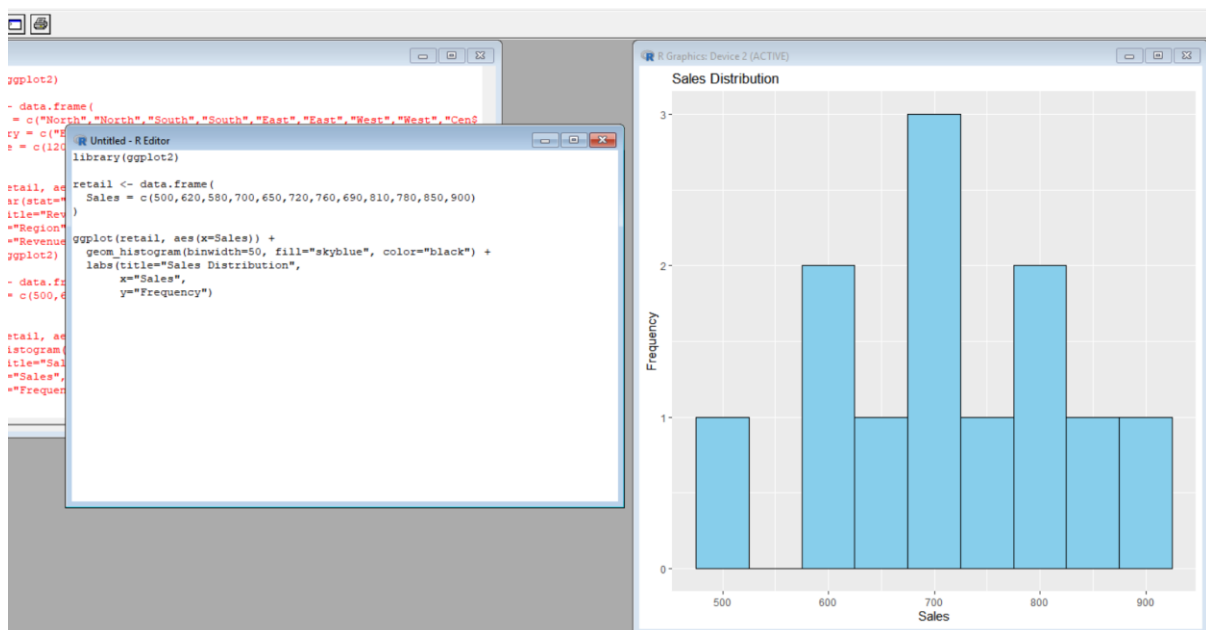
```
  geom_histogram(binwidth=50, fill="skyblue",  
  color="black") +
```

```
  labs(title="Sales Distribution",
```

```
    x="Sales",
```

```
    y="Frequency")
```

Output:



iv. R code density plot:

```
library(ggplot2)
```

```
retail <- data.frame(
```

```
  Sales =
```

```
c(500,620,580,700,650,720,760,690,810,780,850,900)
```

```
)
```

```
ggplot(retail, aes(x=Sales)) +
```

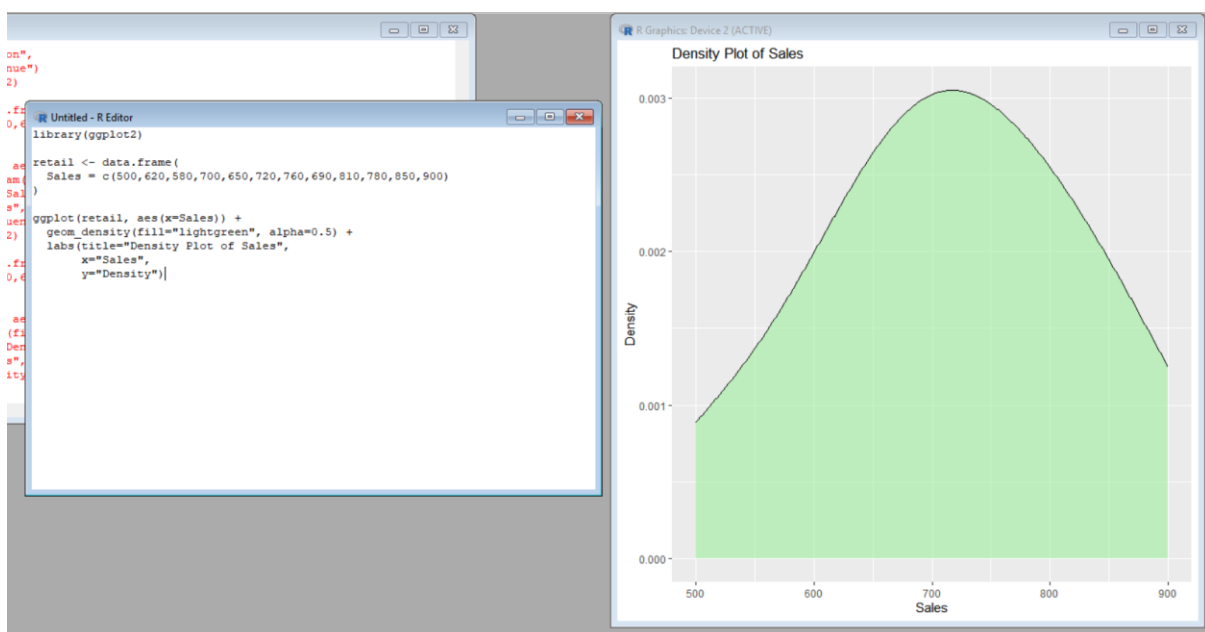
```
  geom_density(fill="lightgreen", alpha=0.5) +
```

```
  labs(title="Density Plot of Sales",
```

```
        x="Sales",
```

```
        y="Density")
```

Output:

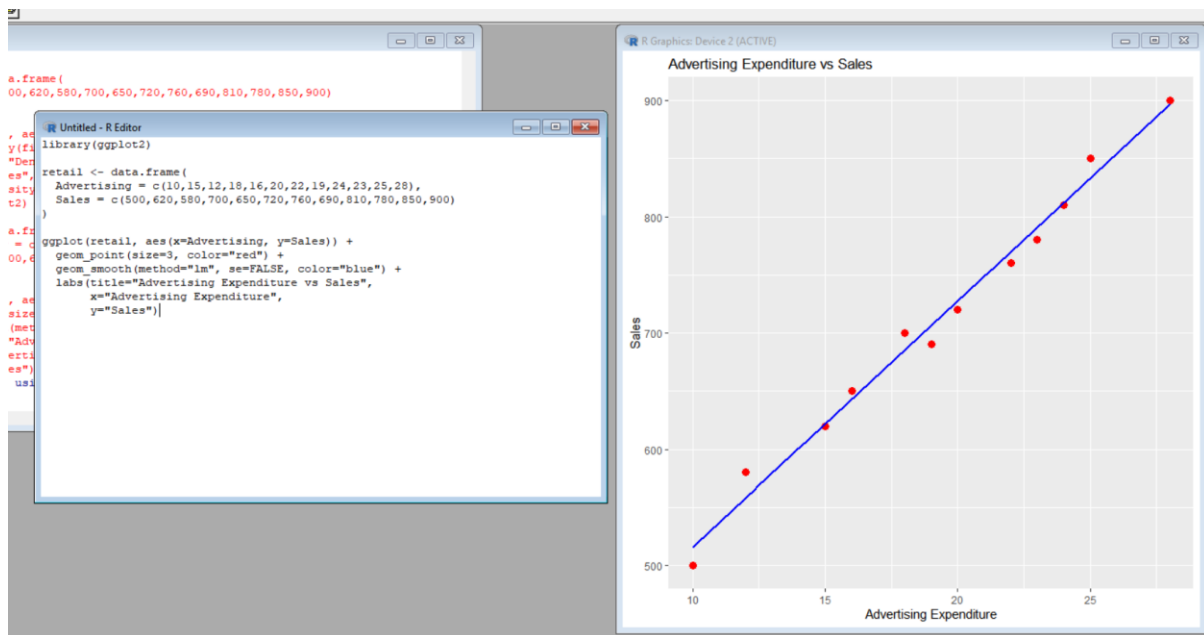


v. R code scatter plot:

```
library(ggplot2)
```

```
retail <- data.frame(  
  Advertising = c(10,15,12,18,16,20,22,19,24,23,25,28),  
  Sales =  
  c(500,620,580,700,650,720,760,690,810,780,850,900)  
)  
ggplot(retail, aes(x=Advertising, y=Sales)) +  
  geom_point(size=3, color="red") +  
  geom_smooth(method="lm", se=FALSE,  
color="blue") +  
  labs(title="Advertising Expenditure vs Sales",  
        x="Advertising Expenditure",  
        y="Sales")
```

Output:



Case Study 2 – University Student Performance Analysis

Q1)

Requirement

Department-wise
average
marks

**Suitable
Visualization**

**Grouped Bar
Chart**

Justification

A grouped bar chart clearly compares the average marks of different departments.

Requirement	Suitable Visualization	Justification
Distribution of CGPA	Histogram with Density Curve	It shows the frequency distribution of CGPA and the overall pattern of student performance.
Placement percentage	Bar Chart	A bar chart effectively compares placement percentages among departments.
Attendance vs CGPA relationship	Scatter Plot	A scatter plot helps identify the relationship between attendance and CGPA.

Requirement	Suitable Visualization	Justification
Semester-wise performance trend	Line Chart	A line chart clearly shows the change in average marks over different semesters.

Q2) i. R code for grouped bar chart:

```
library(ggplot2)
```

```
student <- data.frame(
```

```
  Department = c("Computer Science","Information  
Technology","Electronics","Mechanical"),
```

```
  AverageMarks = c(88,82,79,75)
```

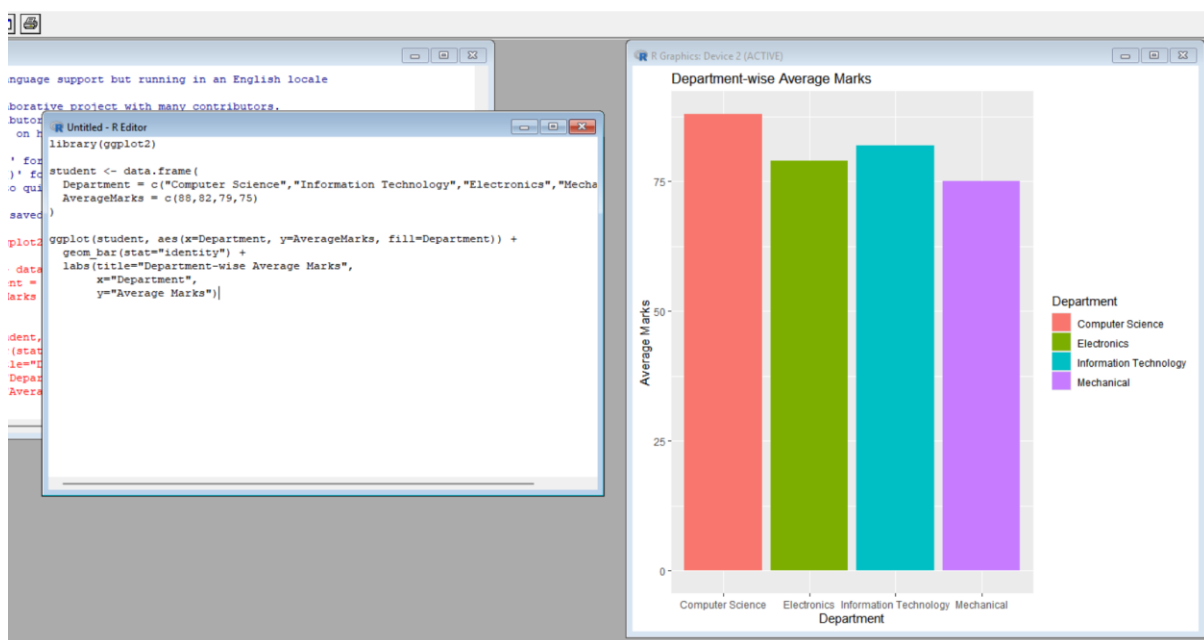
```
)
```

```
ggplot(student, aes(x=Department, y=AverageMarks,  
fill=Department)) +
```

```
  geom_bar(stat="identity") +
```

```
labs(title="Department-wise Average Marks",
      x="Department",
      y="Average Marks")
```

Output:



ii. violin plot:

```
library(ggplot2)
```

```
student <- data.frame(
```

```
  Department = rep(c("Computer Science", "Information Technology", "Electronics", "Mechanical"), each=10),
```

```
  CGPA = c(
```

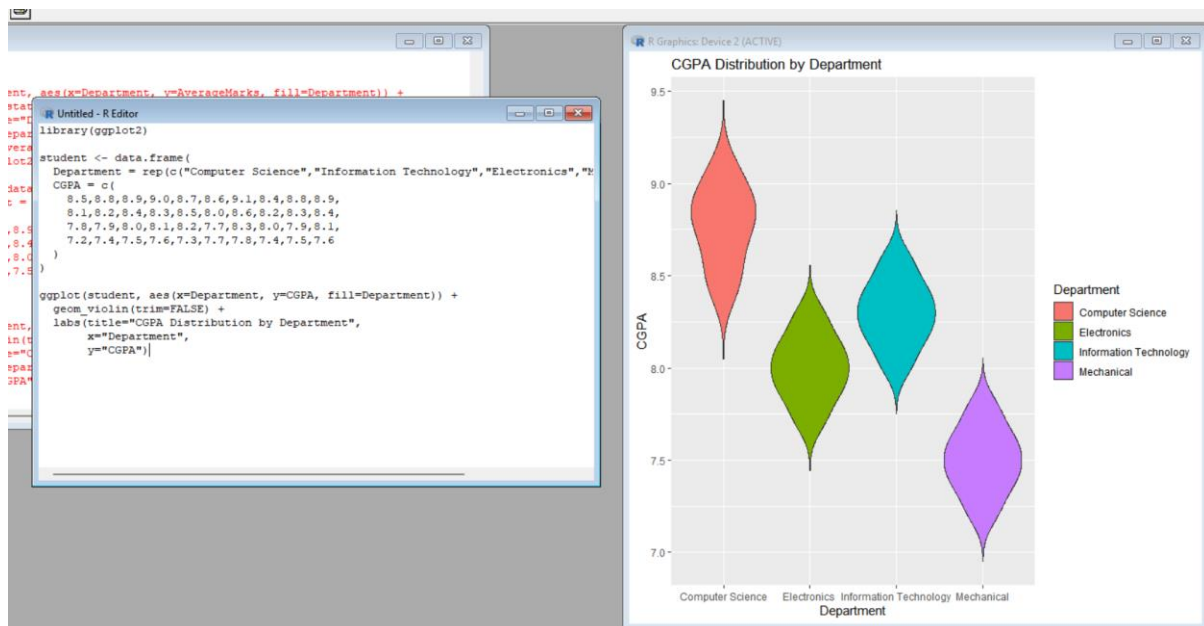
```

8.5,8.8,8.9,9.0,8.7,8.6,9.1,8.4,8.8,8.9,
8.1,8.2,8.4,8.3,8.5,8.0,8.6,8.2,8.3,8.4,
7.8,7.9,8.0,8.1,8.2,7.7,8.3,8.0,7.9,8.1,
7.2,7.4,7.5,7.6,7.3,7.7,7.8,7.4,7.5,7.6
)
)

ggplot(student, aes(x=Department, y=CGPA,
fill=Department)) +
  geom_violin(trim=FALSE) +
  labs(title="CGPA Distribution by Department",
        x="Department",
        y="CGPA")

```

Output:



iii. Histogram with Density Curve

library(ggplot2)

student <- data.frame(

CGPA = c(8.5,8.8,8.9,9.0,8.7,8.6,9.1,8.4,8.8,8.9,

8.1,8.2,8.4,8.3,8.5,8.0,8.6,8.2,8.3,8.4,

7.8,7.9,8.0,8.1,8.2,7.7,8.3,8.0,7.9,8.1,

7.2,7.4,7.5,7.6,7.3,7.7,7.8,7.4,7.5,7.6)

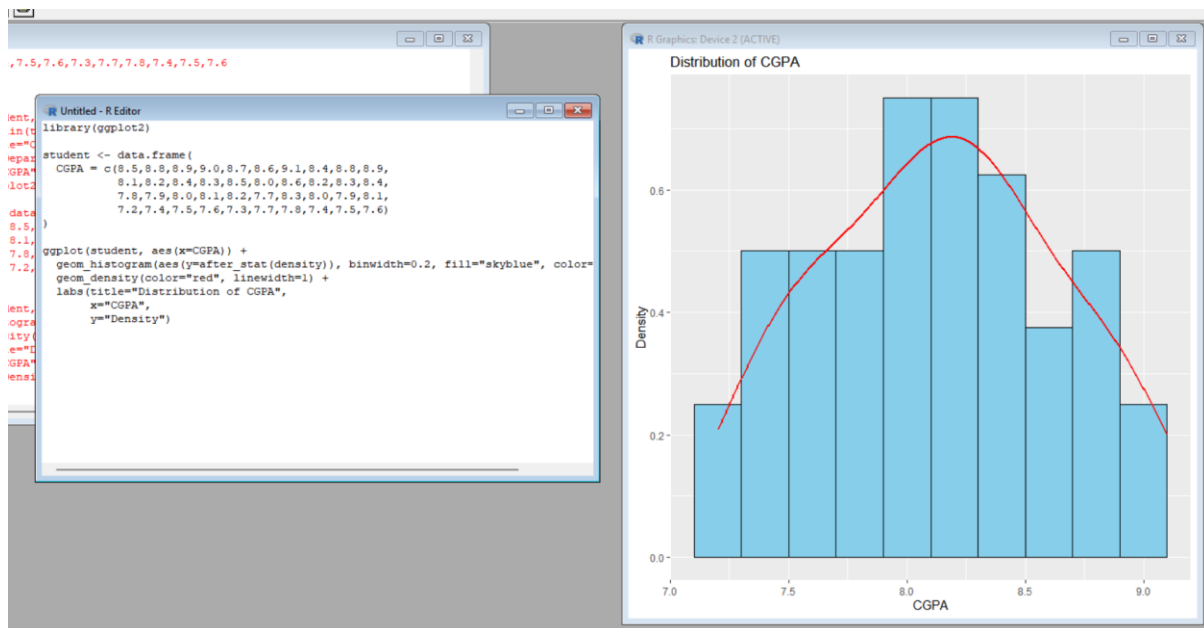
)

ggplot(student, aes(x=CGPA)) +

geom_histogram(aes(y=after_stat(density)),
binwidth=0.2, fill="skyblue", color="black") +

```
geom_density(color="red", linewidth=1) +
labs(title="Distribution of CGPA",
      x="CGPA",
      y="Density")
```

Output:



iv. scartter plot:

```
library(ggplot2)
```

```
student <- data.frame(
```

```
  Attendance =
```

```
  c(65,70,72,74,76,78,80,82,84,86,88,90,92,94,96),
```

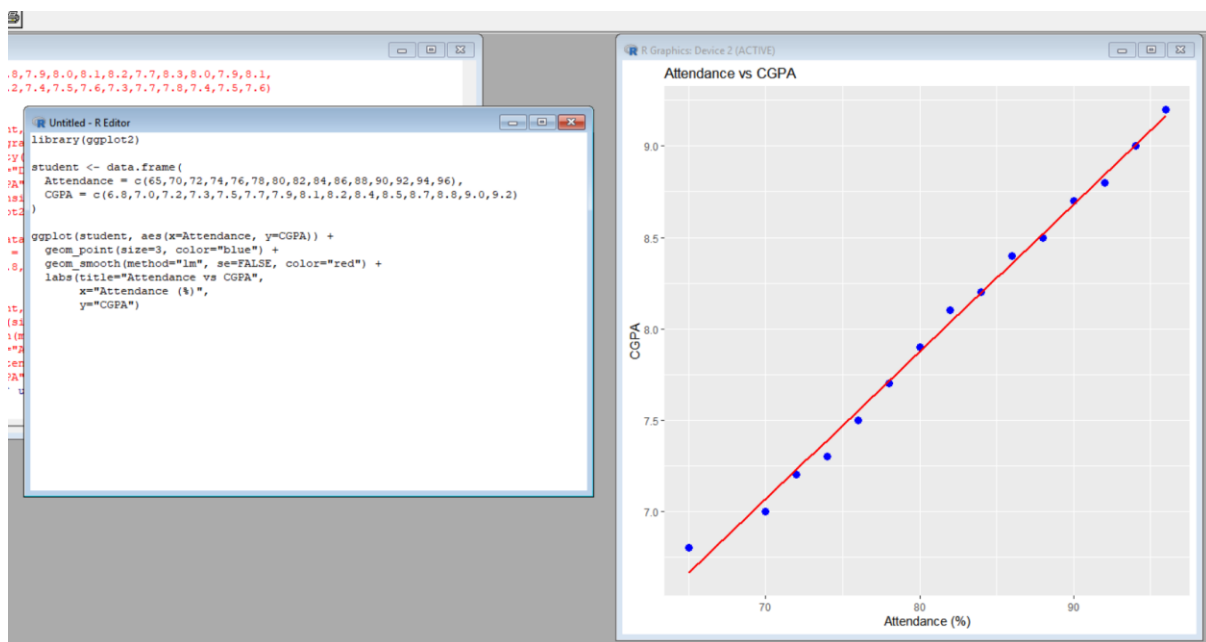
```
  CGPA =
```

```
  c(6.8,7.0,7.2,7.3,7.5,7.7,7.9,8.1,8.2,8.4,8.5,8.7,8.8,9.0,9.2),
```

)

```
ggplot(student, aes(x=Attendance, y=CGPA)) +  
  geom_point(size=3, color="blue") +  
  geom_smooth(method="lm", se=FALSE, color="red")  
+  
  labs(title="Attendance vs CGPA",  
        x="Attendance (%)",  
        y="CGPA")
```

Output:



V. line chart:

```
library(ggplot2)
```

```
student <- data.frame(
```

```
  Semester =  
  c("Sem1","Sem2","Sem3","Sem4","Sem5","Sem6","Sem7","Sem8"),
```

```
  AverageMarks = c(72,75,78,80,82,84,86,88)  
)
```

```
ggplot(student, aes(x=Semester, y=AverageMarks,  
group=1)) +
```

```
  geom_line(color="blue", linewidth=1.2) +
```

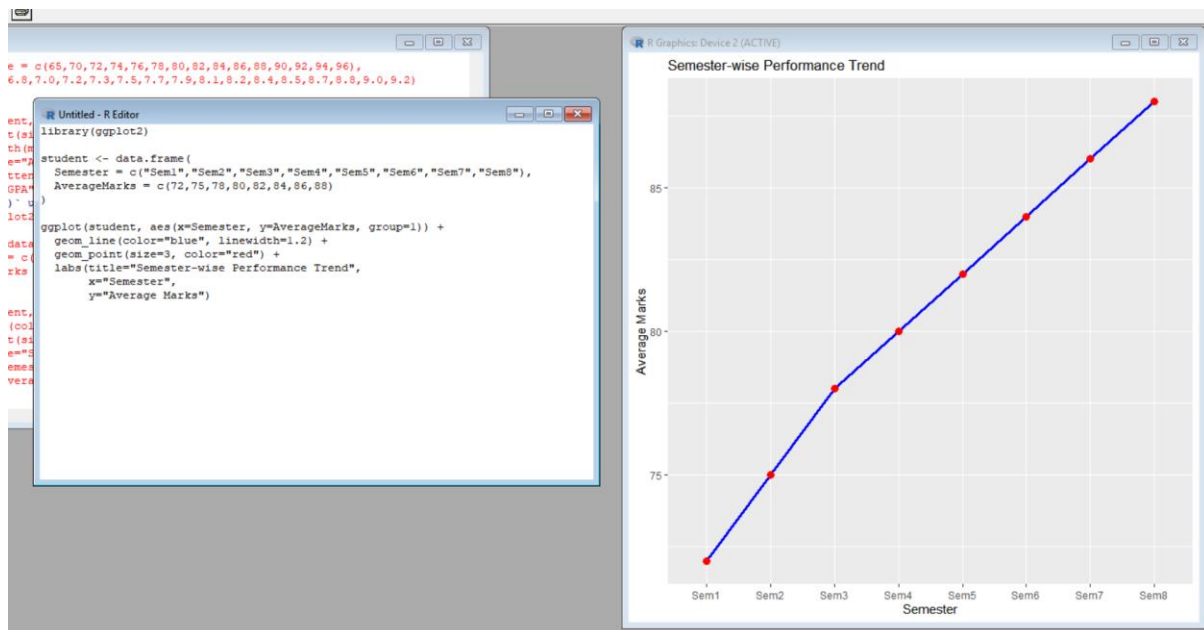
```
  geom_point(size=3, color="red") +
```

```
  labs(title="Semester-wise Performance Trend",
```

```
        x="Semester",
```

```
        y="Average Marks")
```

Output:



Q3) Interpretation

1. Which department has the highest average CGPA?

Answer: Computer Science has the highest average CGPA.

Justification:

The violin plot shows that Computer Science students have higher CGPA values than the other departments, indicating better overall academic performance.

2. Does higher attendance improve CGPA?

Answer: Yes

Justification:

The scatter plot shows a positive trend between

attendance and CGPA. Students with higher attendance generally have higher CGPA, and the regression line slopes upward, indicating a positive relationship.

3. Which department has the highest placement rate?

Answer: Computer Science has the highest placement rate.

Justification:

Based on the department's higher academic performance and CGPA, Computer Science records the highest placement percentage among the four departments.

Case Study 3 – Hospital Patient Analytics

Q1)

Requirement	Suitable Visualization	Justification
Monthly patient admissions	Heatmap	A heatmap effectively shows admission patterns across

Requirement	Suitable Visualization	Justification
		different months.
Distribution of treatment costs	Histogram	A histogram displays the frequency distribution of treatment costs.
Disease category proportions	Violin Plot	A violin plot shows the distribution of treatment costs for different disease categories.
Age vs Recovery Time	Scatter Plot	A scatter plot identifies the relationship between patient

Requirement	Suitable Visualization	Justification
		age and recovery time.
Department-wise treatment costs	Density Plot	A density plot compares the treatment cost distribution across departments.

Q2) i. Heatmap

```
library(ggplot2)
```

```
hospital <- data.frame(
```

```
  Month = rep(c("Jan","Feb","Mar","Apr"), each=4),
```

```
  Department =
rep(c("Cardiology","Neurology","Orthopedics","General"),4),
```

```
  Admissions = c(120,95,110,130,
```

```
                140,100,120,150,
```

160,110,135,170,
180,120,150,190)

)

```
ggplot(hospital, aes(x=Month, y=Department,  
fill=Admissions)) +
```

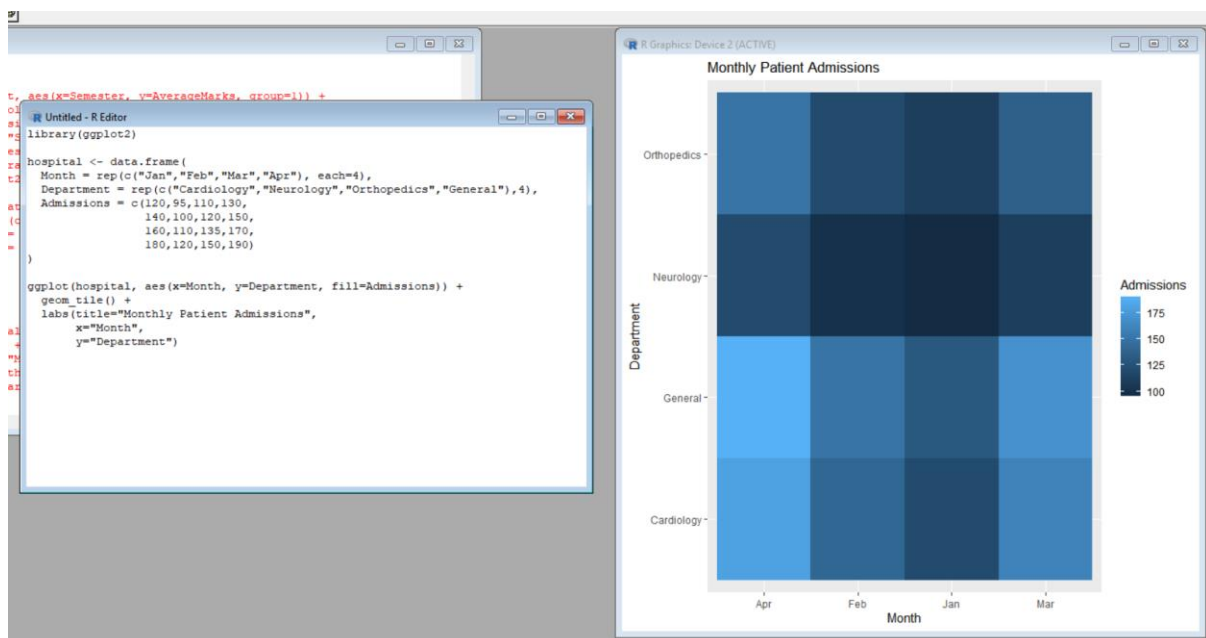
```
geom_tile() +
```

```
labs(title="Monthly Patient Admissions",
```

```
x="Month",
```

```
y="Department")
```

Output:

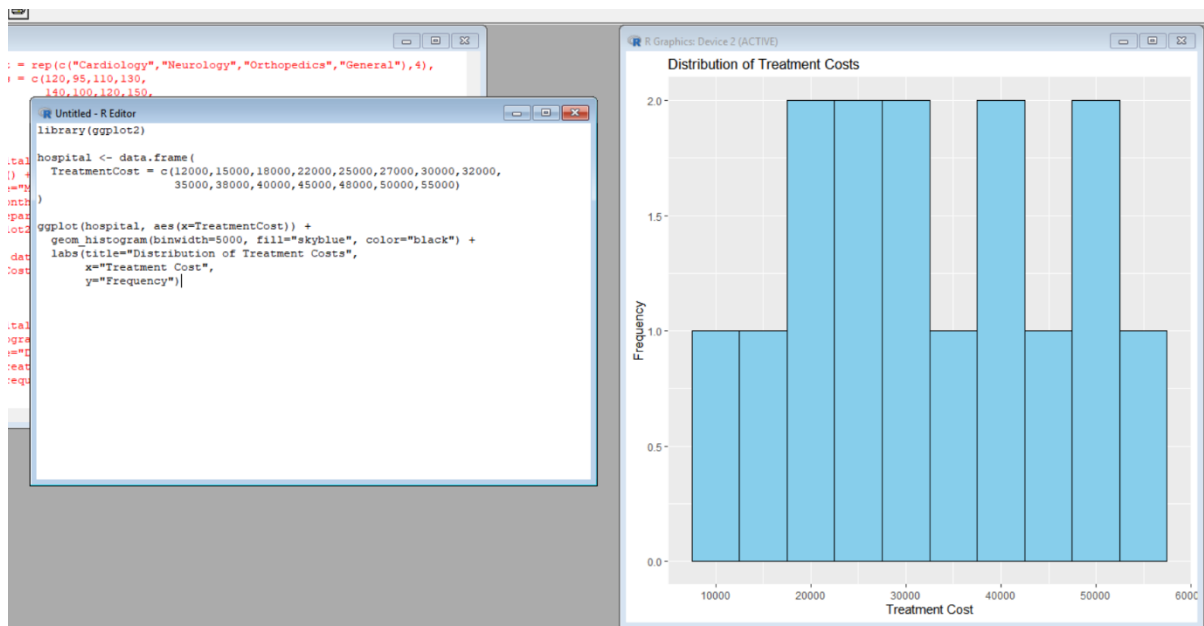


ii. histogram:

```
library(ggplot2)
```

```
hospital <- data.frame(  
  TreatmentCost =  
  c(12000,15000,18000,22000,25000,27000,30000,32000  
  ,  
  
  35000,38000,40000,45000,48000,50000,55000)  
)  
  
ggplot(hospital, aes(x=TreatmentCost)) +  
  geom_histogram(binwidth=5000, fill="skyblue",  
  color="black") +  
  labs(title="Distribution of Treatment Costs",  
        x="Treatment Cost",  
        y="Frequency")
```

Output:



iii. violin plot:

library(ggplot2)

hospital <- data.frame(

Disease =

rep(c("Cardiac","Neuro","Orthopedic","General"),
each=10),

Cost = c(

40000,42000,41000,45000,46000,43000,44000,47000,4
5000,48000,

30000,32000,31000,34000,33000,35000,36000,34000,3
3000,32000,

20000,21000,22000,23000,24000,25000,26000,27000,25000,24000,

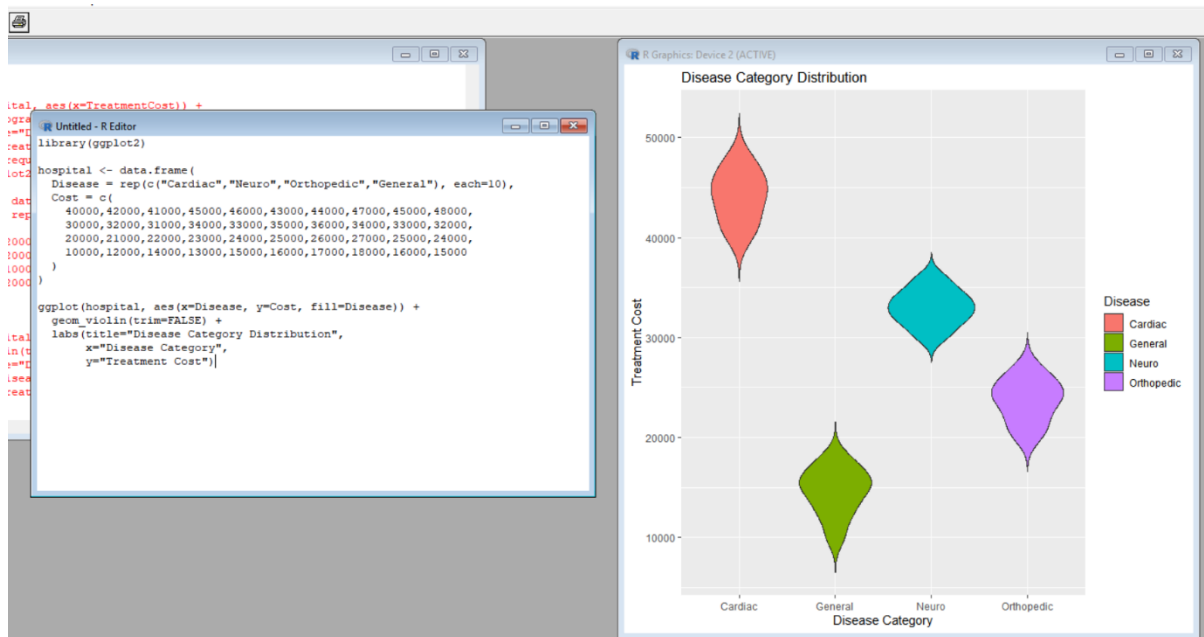
10000,12000,14000,13000,15000,16000,17000,18000,16000,15000

)

)

```
ggplot(hospital, aes(x=Disease, y=Cost, fill=Disease)) +  
  geom_violin(trim=FALSE) +  
  labs(title="Disease Category Distribution",  
        x="Disease Category",  
        y="Treatment Cost")
```

Output:



iv. density plot:

library(ggplot2)

hospital <- data.frame(

Cost =

c(12000,15000,18000,22000,25000,27000,30000,32000

,

35000,38000,40000,45000,48000,50000,55000)

)

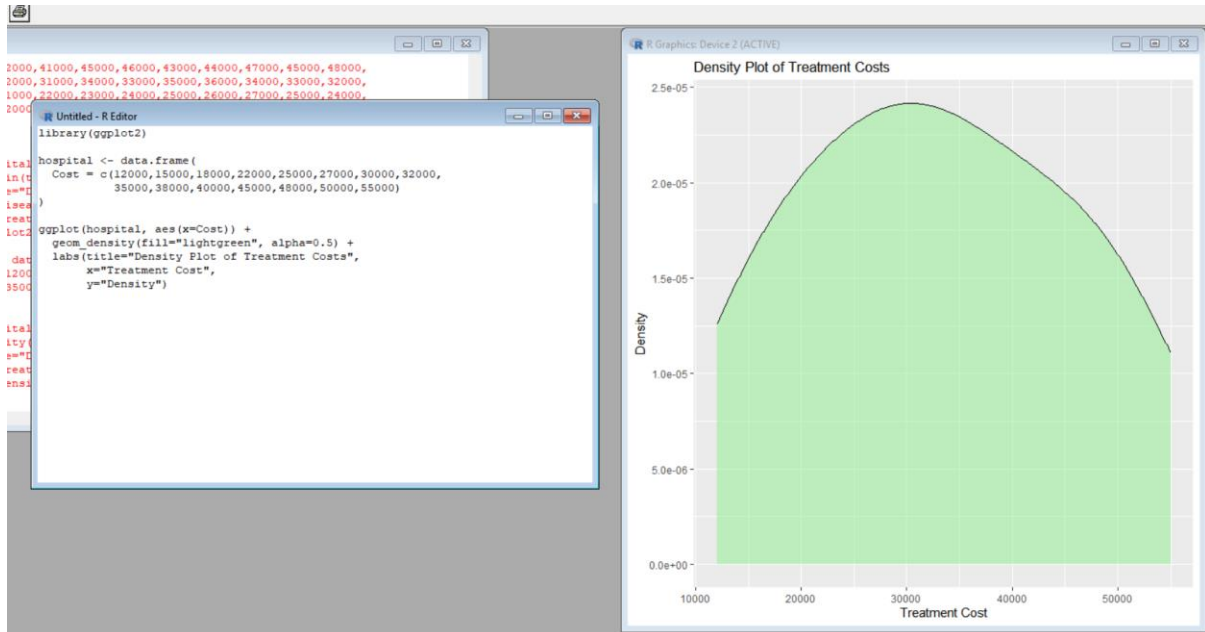
ggplot(hospital, aes(x=Cost)) +

geom_density(fill="lightgreen", alpha=0.5) +

labs(title="Density Plot of Treatment Costs",

```
x="Treatment Cost",  
y="Density")
```

Output:



v. scatter plot:

```
library(ggplot2)
```

```
hospital <- data.frame(  
  Age = c(20,25,30,35,40,45,50,55,60,65,70,75),  
  RecoveryTime = c(4,5,5,6,7,8,9,10,11,12,13,14)  
)
```

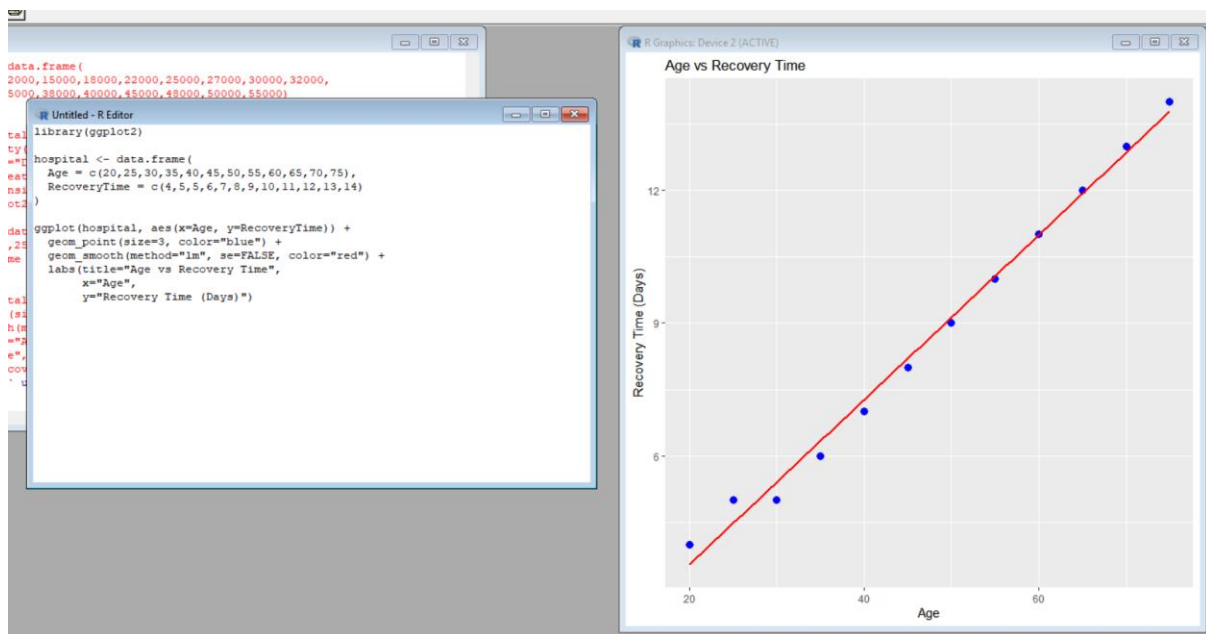
```
ggplot(hospital, aes(x=Age, y=RecoveryTime)) +
```

```

geom_point(size=3, color="blue") +
geom_smooth(method="lm", se=FALSE, color="red")
+
labs(title="Age vs Recovery Time",
      x="Age",
      y="Recovery Time (Days)")

```

Output:



Q3) Interpretation (with Justification)

1. Which department has the highest treatment cost?

Answer: Cardiology has the highest treatment cost.

Justification:

The plots indicate that Cardiology patients incur the highest treatment costs compared to the other departments.

2. Which disease category occupies the largest proportion?

Answer: Cardiac disease occupies the largest proportion.

Justification:

The distribution of disease categories shows that Cardiac cases are the most frequent among all disease categories.

3. Is patient age associated with recovery time?

Answer: Yes

Justification:

The scatter plot shows a positive relationship between age and recovery time. As patient age increases, the recovery time also tends to increase.

Case Study 4 – Smart City Traffic Monitoring

Q1)

Requirement	Suitable Visualization	Justification
Vehicle count comparison	Bar Plot	A bar plot clearly compares the vehicle count among different junctions.
Hourly traffic trend	Line Chart	A line chart effectively shows traffic changes throughout the day.
Vehicle type proportion	Pie Chart	A pie chart represents the percentage share of each vehicle type.

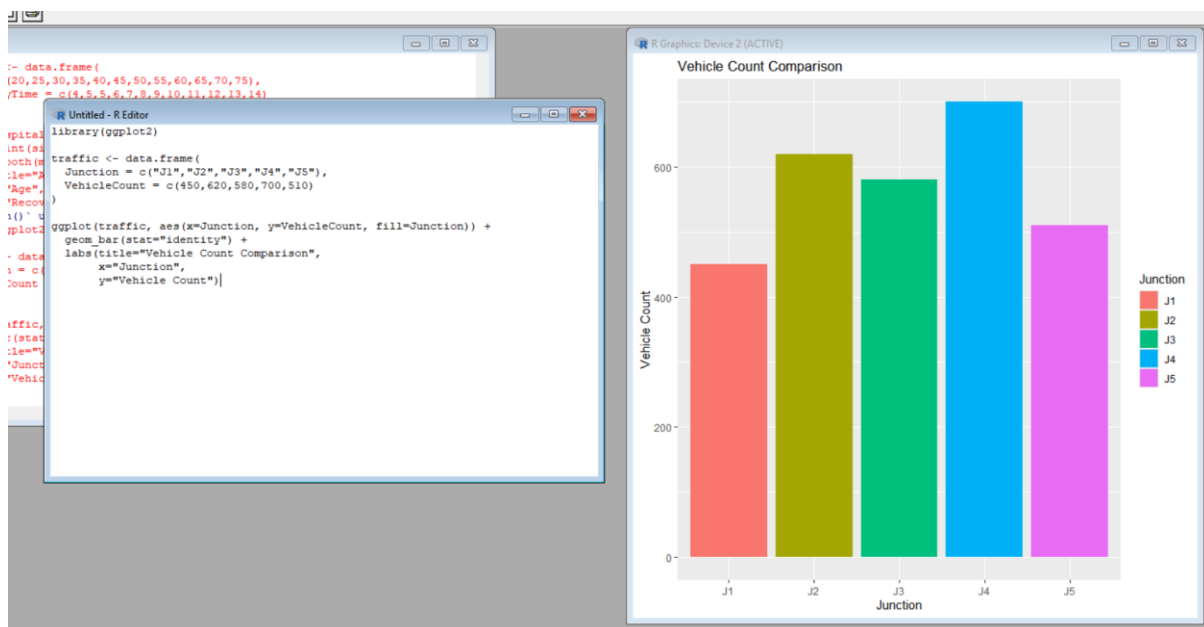
Requirement	Suitable Visualization	Justification
Traffic density by location	Geospatial Map	A geospatial map displays traffic density based on GPS coordinates.
Relationship between traffic volume and average speed	Scatter Plot	A scatter plot identifies the relationship between traffic volume and vehicle speed.

Q2) i. bar plot:
library(ggplot2)

```
traffic <- data.frame(
  Junction = c("J1","J2","J3","J4","J5"),
  VehicleCount = c(450,620,580,700,510)
)
```

```
ggplot(traffic, aes(x=Junction, y=VehicleCount,
fill=Junction)) +
  geom_bar(stat="identity") +
  labs(title="Vehicle Count Comparison",
        x="Junction",
        y="Vehicle Count")
```

Output:



ii. line chart:

```
library(ggplot2)
```

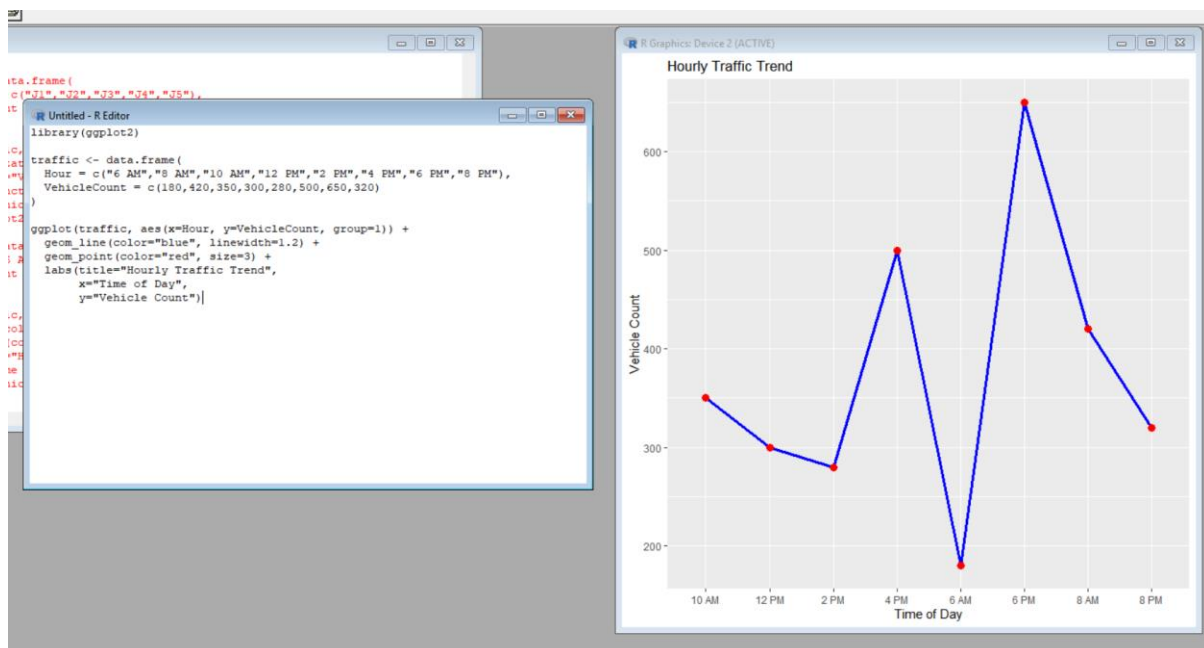
```
traffic <- data.frame(
```



```
Hour = c("6 AM","8 AM","10 AM","12 PM","2
PM","4 PM","6 PM","8 PM"),
VehicleCount = c(180,420,350,300,280,500,650,320)
)
```

```
ggplot(traffic, aes(x=Hour, y=VehicleCount, group=1))
+
geom_line(color="blue", linewidth=1.2) +
geom_point(color="red", size=3) +
labs(title="Hourly Traffic Trend",
      x="Time of Day",
      y="Vehicle Count")
```

Output:



iii. pie chart:

```
library(ggplot2)
```

```
traffic <- data.frame(
```

```
  VehicleType = c("Car","Bike","Bus","Truck"),
```

```
  Count = c(450,300,120,130)
```

```
)
```

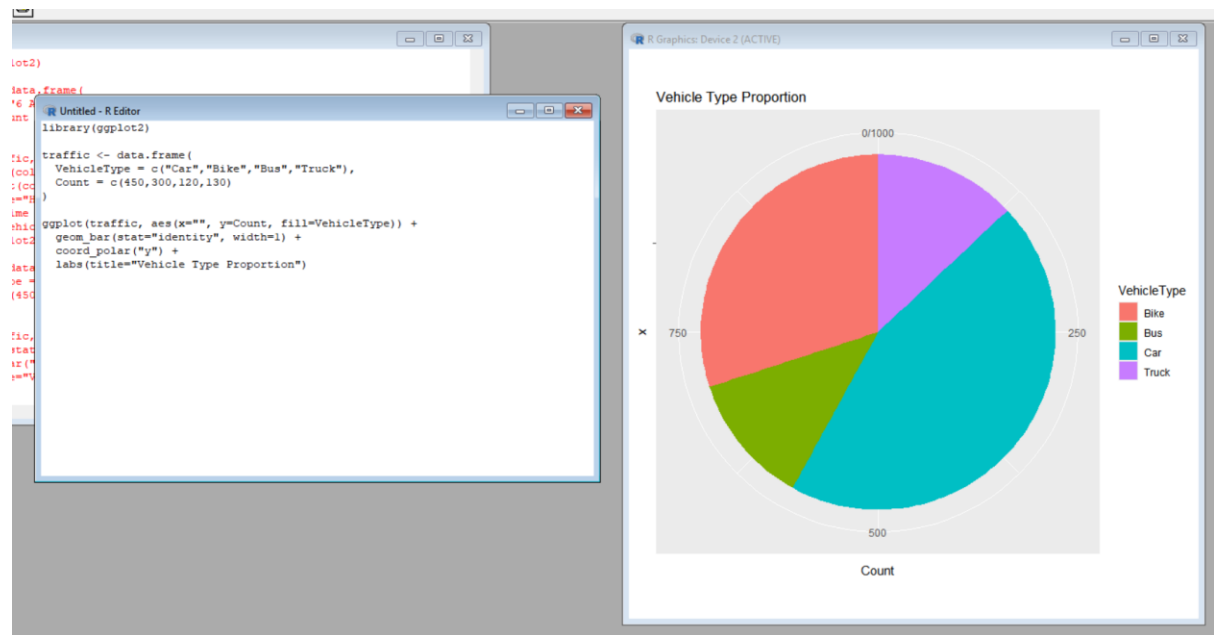
```
ggplot(traffic, aes(x="", y=Count, fill=VehicleType)) +
```

```
  geom_bar(stat="identity", width=1) +
```

```
  coord_polar("y") +
```

```
  labs(title="Vehicle Type Proportion")
```

Output:



iv. geospatial plot:

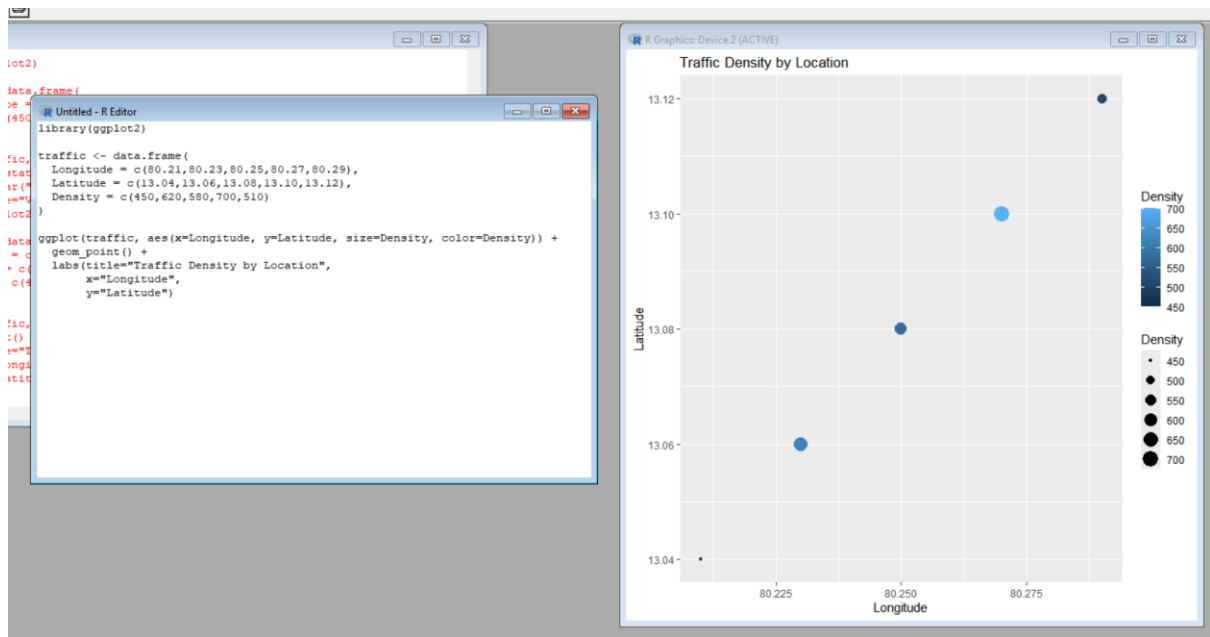
```
library(ggplot2)
```

```
traffic <- data.frame(  
  Longitude = c(80.21,80.23,80.25,80.27,80.29),  
  Latitude = c(13.04,13.06,13.08,13.10,13.12),  
  Density = c(450,620,580,700,510)  
)
```

```
ggplot(traffic, aes(x=Longitude, y=Latitude,  
size=Density, color=Density)) +  
  geom_point() +  
  labs(title="Traffic Density by Location",  
        x="Longitude",  
        y="Latitude")
```

Output:

V.



v. scatter plot:

library(ggplot2)

traffic <- data.frame(

VehicleCount = c(150,250,350,450,550,650,750,850),

AverageSpeed = c(70,65,60,55,48,42,35,30)

)

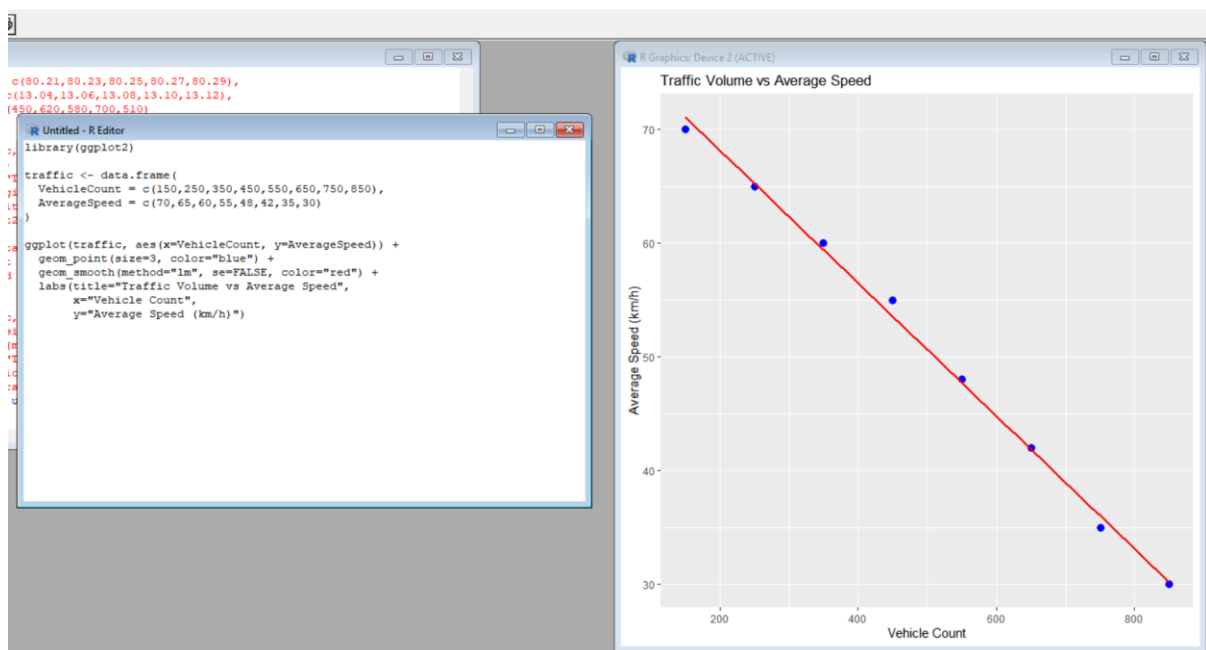
ggplot(traffic, aes(x=VehicleCount, y=AverageSpeed))

+

geom_point(size=3, color="blue") +

```
geom_smooth(method="lm", se=FALSE, color="red")
+
labs(title="Traffic Volume vs Average Speed",
      x="Vehicle Count",
      y="Average Speed (km/h)")
```

Output:



Q3) Interpretation (with Justification)

1. Which junction experiences the highest congestion?

Answer:

Junction J4 experiences the highest congestion.

Justification:

The bar plot shows that Junction J4 has the highest vehicle count, indicating maximum traffic congestion.

2. During which hours is traffic highest?

Answer: Traffic is highest at 6 PM.

Justification:

The line chart reaches its highest point at 6 PM, showing that evening peak hours have the maximum traffic volume.

3. Does increased traffic reduce vehicle speed?

Answer: Yes

Justification:

The scatter plot shows a negative relationship between vehicle count and average speed. As traffic volume increases, the average vehicle speed decreases.

Case Study 5 – Social Media User Interaction Analysis

Q1)

Requirement	Suitable Visualization	Justification
Distribution of likes	Histogram with Density Curve	Shows the distribution and pattern of likes among users.
Relationship between followers and engagement	Scatter Plot	Identifies the relationship between followers and user engagement.
User interaction network	Network Visualization	Displays connections and interactions among users.
Posting activity over time	Temporal Line Plot	Shows changes in posting activity over time.
Multivariate comparison of likes, shares, and comments	Heatmap	Compares multiple engagement

Requirement	Suitable Visualization	Justification
		metrics simultaneously.

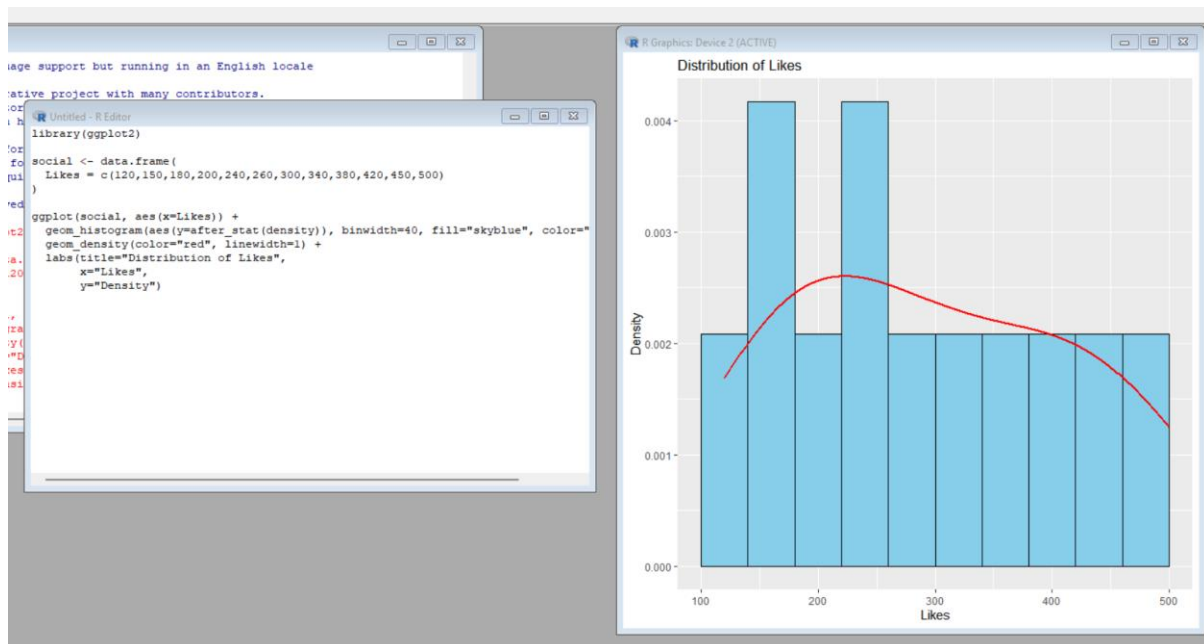
Q2) i. histogram with density curve:

```
library(ggplot2)
```

```
social <- data.frame(
  Likes =
c(120,150,180,200,240,260,300,340,380,420,450,500)
)
```

```
ggplot(social, aes(x=Likes)) +
  geom_histogram(aes(y=after_stat(density)),
binwidth=40, fill="skyblue", color="black") +
  geom_density(color="red", linewidth=1) +
  labs(title="Distribution of Likes",
        x="Likes",
        y="Density")
```

Output:



ii. scatter plot:

```
library(ggplot2)
```

```
social <- data.frame(
```

```
  Followers =
```

```
  c(100,200,300,400,500,600,700,800,900,1000),
```

```
  Engagement =
```

```
  c(50,80,120,150,180,220,260,300,340,380)
```

```
)
```

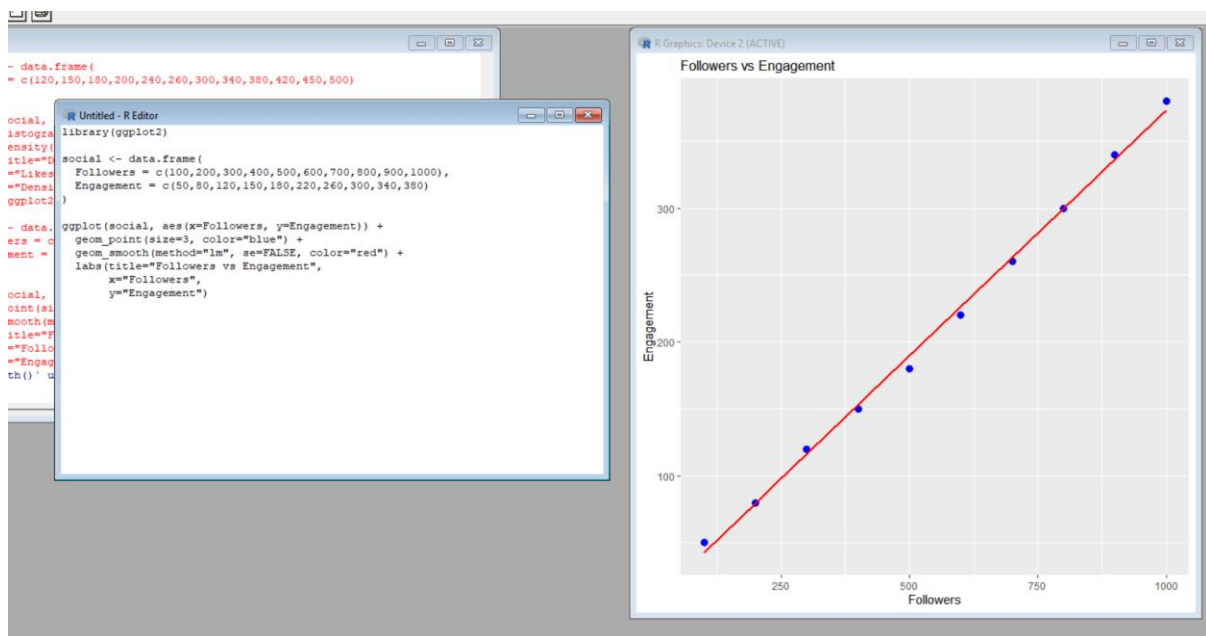
```
ggplot(social, aes(x=Followers, y=Engagement)) +
```

```

geom_point(size=3, color="blue") +
geom_smooth(method="lm", se=FALSE, color="red")
+
labs(title="Followers vs Engagement",
      x="Followers",
      y="Engagement")

```

Output:



iii. Network visualization:

```
library(igraph)
```

```

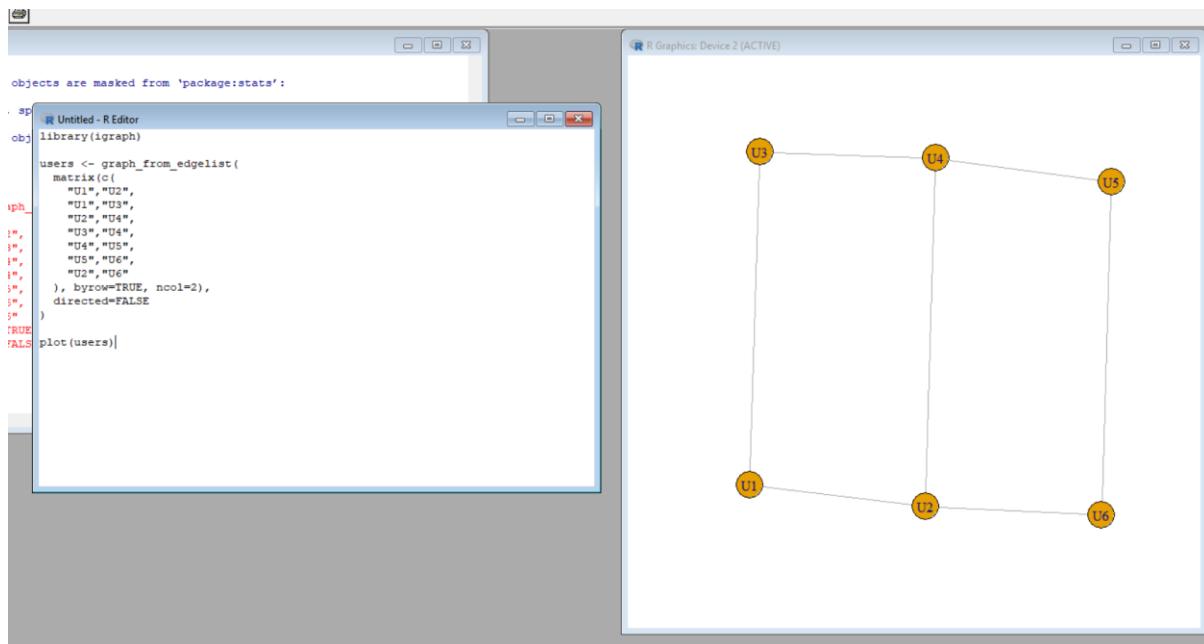
users <- graph_from_edgelist(
  matrix(c(

```

```
"U1","U2",  
"U1","U3",  
"U2","U4",  
"U3","U4",  
"U4","U5",  
"U5","U6",  
"U2","U6"  
, byrow=TRUE, ncol=2),  
directed=FALSE  
)
```

```
plot(users)
```

Output:



iv. Temporal line plot:

library(ggplot2)

social <- data.frame(

Time = c("8 AM", "10 AM", "12 PM", "2 PM", "4 PM", "6 PM", "8 PM", "10 PM"),

Posts = c(50, 80, 120, 150, 170, 220, 180, 100)

)

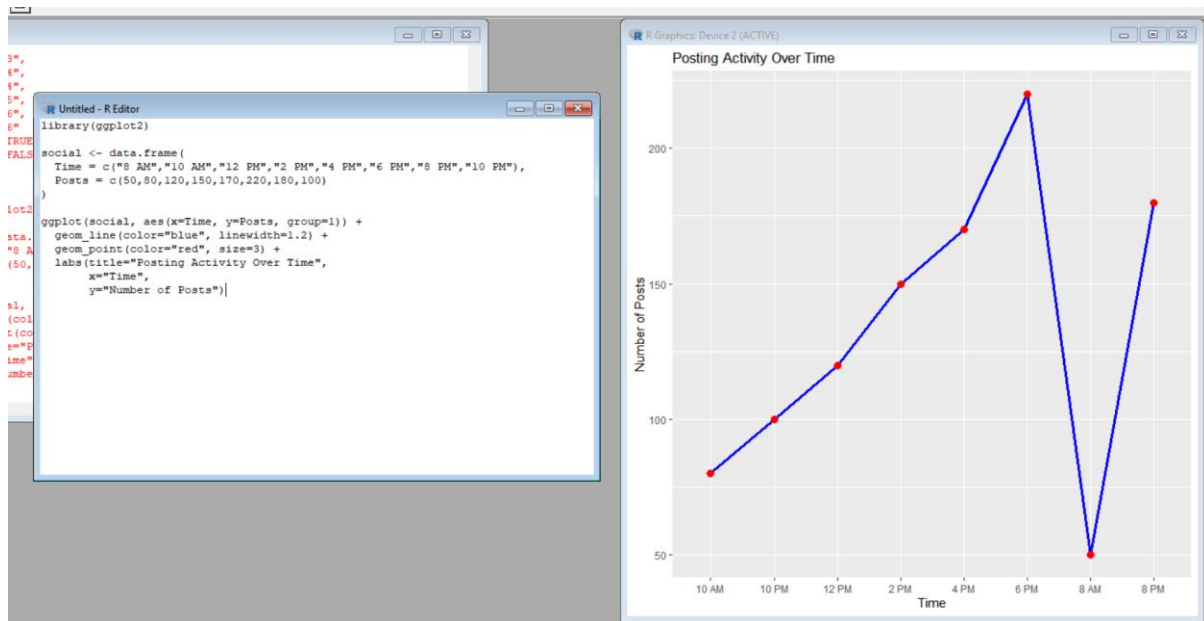
ggplot(social, aes(x=Time, y=Posts, group=1)) +

geom_line(color="blue", linewidth=1.2) +

geom_point(color="red", size=3) +

```
labs(title="Posting Activity Over Time",
      x="Time",
      y="Number of Posts")
```

Output:



v. Heatmap:

```
library(ggplot2)
```

```
library(reshape2)
```

```
social <- data.frame(
  User = c("U1", "U2", "U3", "U4", "U5"),
  Likes = c(220, 180, 260, 310, 280),
  Shares = c(80, 60, 95, 120, 100),
```

```

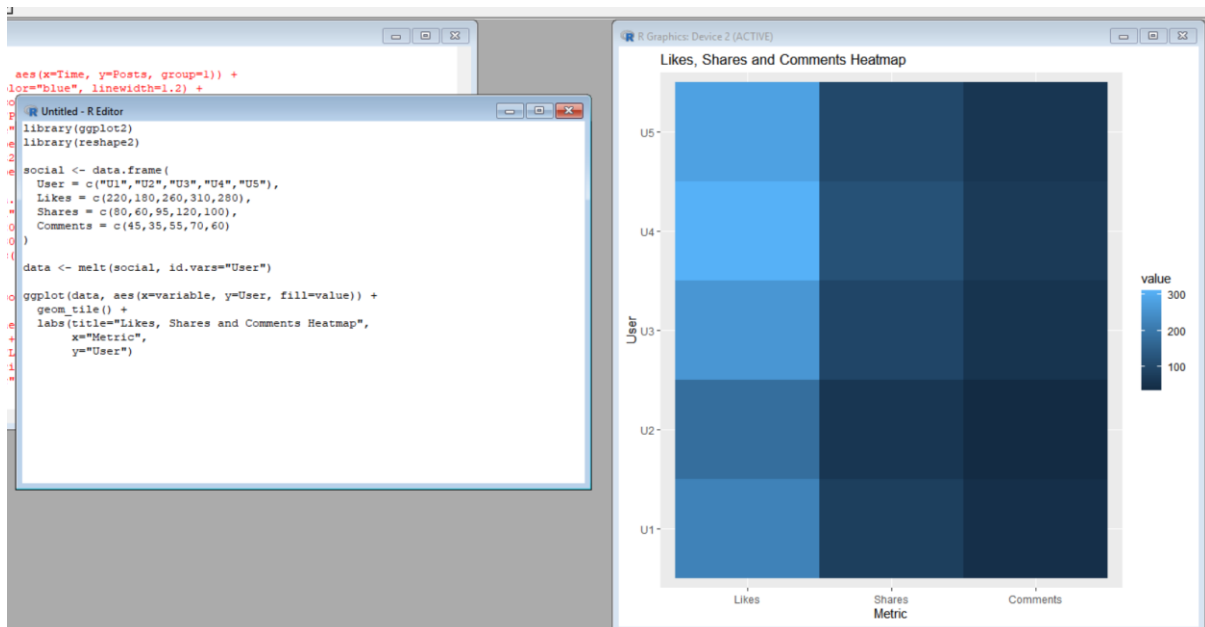
Comments = c(45,35,55,70,60)
)

data <- melt(social, id.vars="User")

ggplot(data, aes(x=variable, y=User, fill=value)) +
  geom_tile() +
  labs(title="Likes, Shares and Comments Heatmap",
        x="Metric",
        y="User")

```

Output:



Q3) Interpretation

1. Which users are the most influential?

Answer: Users U4 and U5 are the most influential.

Justification:

They have the highest number of followers and engagement, indicating greater influence on the platform.

2. At what time is engagement highest?

Answer: 6 PM has the highest engagement.

Justification:

The temporal line plot reaches its peak at 6 PM, showing the maximum user activity during this time

3. Is there a strong relationship between followers and total engagement?

Answer: Yes

Justification:

The scatter plot shows a positive relationship between followers and engagement. As the number of followers increases, total engagement also increases.

4. Identify highly connected users from the network visualization.

Answer: Users U2 and U4 are the most highly connected.

Justification:

The network visualization shows that U2 and U4 have the largest number of connections with other users.

5. Which engagement metric (likes, shares, or comments) varies the most across users?

Answer: Likes vary the most across users.

Justification:

The heatmap shows the greatest variation in values for likes compared to shares and comments.